

The molecular basis of nutritional intervention in multiple sclerosis: a narrative review.

It is commonly accepted that nutrition is one of the possible environmental factors involved in the pathogenesis of multiple sclerosis (MS), but its role as complementary MS treatment is unclear and largely disregarded. At present, MS therapy is not associated to a particular diet, probably due to lack of information on the effects of nutrition on the disease. To overcome the distrust of the usefulness of dietary control in MS and to encourage nutritional interventions in the course of the disease, it is necessary to assess the nature and the role of bioactive dietary molecules and their targets, and establish how a dietary control can influence cell metabolism and improve the wellness of MS patients. The aim of this review is to provide a rationale for a nutritional intervention in MS by evaluating at the molecular level the effects of dietary molecules on the inflammatory and autoimmune processes involved in the disease. Present data reveal that healthy dietary molecules have a pleiotropic role and are able to change cell metabolism from anabolism to catabolism and down-regulate inflammation by interacting with enzymes, nuclear receptors and transcriptional factors. The control of gut dysbiosis and the combination of hypo-caloric, low-fat diets with specific vitamins, oligoelements and dietary integrators, including fish oil and polyphenols, may slow-down the progression of the disease and ameliorate the wellness of MS patients. Copyright © 2011 Elsevier Ltd. All rights reserved.